1-4 $\alpha = 30^\circ$, $\rho = 0.8 g/cm^3$ $L = 200mm$ $B = 0.1 MPa$

$p = B \mp \rho g L \sin\alpha$

$= 0.1 \times 10^6 - 0.8 \times 9.8 \times 200 \times 10^{-3} \times 0.5 \times 10^3$

$= 100 - 0.784$

~~$= 100.784$ (kPa)~~

$= 99.216$ (kPa)

1-5 $m = 0.5kg$ $p_1 = 0.7 MPa$ $V_1 = 0.02 m^3$ $V_2 = 0.05 m^3$

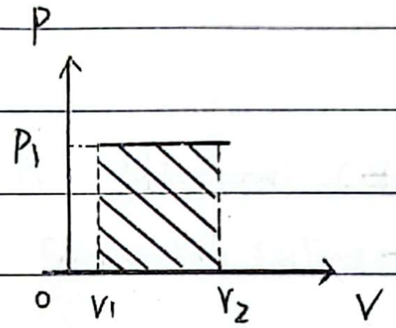
(1). $P =$ 定值

$W = \cancel{\Delta(PV)} \int_{V_1}^{V_2} P dV$

$= P_1(V_2 - V_1)$

$= 0.7 \times 0.03 \times 10^6$

$= 21$ (kJ)



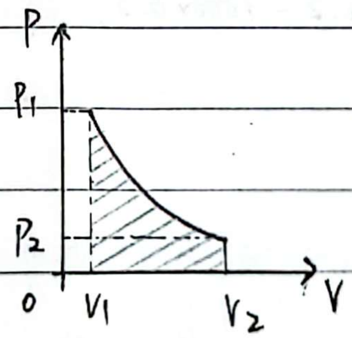

(2). $PV =$ 定值

$W = \int_{V_1}^{V_2} P dV$

$= \int_{0.02}^{0.05} \frac{14 \times 10^3}{V} dV$

$= 14 \times 10^3 \ln V \Big|_{0.02}^{0.05}$

$= 12.83$ (kJ)




习1-6 $\eta = P / mQ_L \times 100\%$

$= 5 \times 10^3 / 19 \times 10^3 \times 3 \times 10^4 \times 100\%$ $\times 3600$

$= 3.16\%$

1-7 电灯产生的热量 $Q_{电} = 40\times2\times3600\times10^{-3} = 288 (kJ/h)$

热泵提供的热 $q_1 = 70000 - Q_{电}$

$= \cancel{69712}\ 69352\ (kJ/h)$

$\varepsilon_2 = q_1/(q_1 - q_2)$

$= \cancel{69712}\ 69352/(69712 - q_2) = 5$

$\therefore q_2 = 55769.6\ (kJ/h)$

驱动热泵的功率 $P = (q_1 - q_2)/3600 = \cancel{15.49}\ (kW)$

$= 3.85 (kW)$

9.28

9.30

2-6 $T_0 = 288K$, $P_0 = 0.1MPa$ $V_0 = 3m^3$ $V = 8.5m^3$ $P = 0.7MPa$

$$\begin{cases} P_0V_0 = n_0RT_0 \\ (P_0+P)V = nRT_0 \\ \cancel{n/n_0 = PV/P_0V_0 = 7\times 8.5\times\frac{1}{3} = 19.8} \\ P_0V = n_1RT_0 \end{cases}$$

$n/n_0 = (P_0+P)V / P_0V_0$

$n_1/n_0 = \frac{V}{V_0}$

$$t = \frac{n-n_1}{n_0} = \frac{(P_0+P)V}{P_0V_0} - \frac{V}{V_0} = \frac{PV}{P_0V_0} = 7\times 8.5\times\frac{1}{3} = 19.8(min)$$

2-7 $t = 300°C$ $P_g = 15.2kPa$. $q = 1.02\times10^5 m^3/h$ $B = 101kPa$

标准状态. $T = 273.15K$. $P = 101.\cancel{3}kPa$

$$\begin{cases} (B+P_g)q = nRt \\ Pq_1 = nRT \end{cases}$$

$\cancel{q_1 = \frac{T}{t}\cdot\frac{P}{(B+P_g)}\cdot q = \frac{273.15\times101}{573.15\times(15.2+101)}\times1.02\times10^5}$

$$q_1 = \frac{T(B+P_g)}{tP}q = \frac{273.15\times(15.2+101)}{573.15\times101}\times1.02\times10^5$$

$$= 5.59\times10^4 (m^3/h)$$

2-14 $t_1 = 25°C$ $t_2 = 250°C$ $V_0 = 3500m^3/h$

(1). $m_0 = V_0 \cdot \rho$

由表2-4, 得 $C_{pm}|_0^{25} = 1.004$ $C_{pm}|_0^{250} = 1.012 + 0.007\times\frac{1}{2} = 1.016$

$\therefore Q = m_0 \cdot t_2 \cdot C_{pm}|_0^{250} - m_0 t_1 C_{pm}|_0^{25}$

$= 3500\times1.2932\times(250\times1.016 - 25\times1.004)$

$= 1.036\times10^6 (kJ/h)$

(2). 由表2-3，得． $a_0 = 28.106$ $a_1 = 1.9665\times10^{-3}$ $a_2 = 4.8023\times10^{-6}$

$a_3 = -1.9661\times10^{-9}$

$$Q = \frac{V_0}{22.4}\int_{T_1}^{T_2}(a_0 + a_1T + a_2T^2 + a_3T^3)$$

$$= \frac{V_0}{22.4}\cdot\left(a_0T + \frac{1}{2}a_1T^2 + \frac{1}{3}a_2T^3 + \frac{1}{4}a_3T^4\right)_{25+273}^{250+273}$$

$$= 1.041\times10^{6}\ (kJ/h)$$

(3). $Q = \frac{V_0}{22.4}\cdot\frac{7}{2}R\cdot\Delta T$

$$= 3500\times\frac{1}{22.4}\times\frac{7}{2}\times8.3145\times225$$

$$= 1.023\times10^{6}\ (kJ/h)$$

2-16 $x_{CO_2} = 0.4$ $x_{N_2} = 0.2$ $x_{O_2} = 0.4$ $t = 50^\circ C$ $p_g = 0.04\,MPa$

$p_b = 750\,mmHg$ ， $V = 4m^3$

(1). 由 $(p_g + p_b)V = nRt$

$$(40 + 750\times13.6\times9.8\times10^{-3})\times4 = n\times8.3145\times323.15$$

$$n = 2.08\times10^{-1}\ (kmol)$$

$$m = n\cdot(x_{CO_2}\cdot M_{CO_2} + x_{N_2}\cdot M_{N_2} + x_{O_2}\cdot M_{O_2})$$

$$= 7.50\ (kg)$$

(2). $V_0 = n\cdot22.4$

$$= 4.66\ (m^3)$$

2-17 $g_{O_2} = 23.2\%$， $g_{N_2} = 76.8\%$

$n_{O_2} = \frac{g_{O_2}/32}{g_{O_2}/32 + g_{N_2}/28} = 20.9\%$ $n_{N_2} = 1 - n_{O_2} = 79.1\%$

$M_{混} = n_{O_2}\cdot32 + n_{N_2}\times28$

$= 28.84\ (g/mol)$

$R = R_0 / M = 288.28 \ (J \cdot K^{-1} \cdot kg^{-1})$

$r_{O_2} = n_{O_2} = 20.9\%$ ; $r_{N_2} = n_{N_2} = 70.1\%$

$\rho = M / 22.4 = 1.29 \ (kg/m^3)$

比体积 $= \frac{1}{\rho} = 0.777 \ (m^3/kg)$

2-18 $r_{CH_4} = 97\%$, $r_{C_2H_6} = 0.6\%$ $r_{C_3H_8} = 0.18\%$ $r_{C_4H_{10}} = 0.18\%$

$r_{CO_2} = 0.2\%$ $r_{N_2} = 1.83\%$

(1). $\rho = \sum r_i M_i / 22.4$

$= \frac{1}{22.4} \times (\cancel{0.7} 0.97 \times 16 + 0.6 \times 10^{-2} \times 30 + 0.18 \times 44 \times 10^{-2} + 0.18 \times 58 \times 10^{-2} + 44 \times 0.2 \times 10^{-2} + 28 \times 1.83 \times 10^{-2})$

$= 0.736 \ (kg/m^3)$

(2). ~~$P_{CH_4}$~~ $P_i = r_i \cdot P_0$, $P_0 = 101.325 \ kPa$.

$\therefore P_{CH_4} = 98.29 \ kPa$ $P_{CO_2} = 2.03 \ kPa$

$P_{C_2H_6} = 0.61 \ kPa$ $P_{N_2} = 1.85 \ kPa$

$P_{C_3H_8} = 1.82 \ kPa$

$P_{C_4H_{10}} = 1.82 \ kPa$

10.10

10.13

3-7 ~~$\Delta U = 1.5(P_2V_2 - P_1V_1)$~~ $m=1.5kg$, $P_1=1000\,kN/m^2$, $V_1=0.2m^3$

~~$1.5\times$~~ $P_2=200\,kN/m^2$ $V_2=1.2m^3$ $p=aV+b$ $u=1.5PV-85$

$\Delta U = 1.5\times(P_2V_2 - P_1V_1)$

$= 1.5\times(200\times1.2 - 1000\times0.2)$

$= 60\ (kJ)$

$\because p = aV+b$

$\therefore \begin{cases} P_1 = aV_1+b \\ P_2 = aV_2+b \end{cases} \Rightarrow \begin{cases} a=-800 \\ b=1160 \end{cases}$

$\therefore P = -800V+1160$

$W = \int_1^2 P\,dV$

$= \int_{0.2}^{1.2}(-800V+1160)\,dV$

$= (-400V^2+1160V)\Big|_{0.2}^{1.2}$

$= 600\ (kJ)$

$-560+1160$

$=600$

$Q = \Delta U + W = 60+600 = 660\ (kJ)$

3-8 $P_1 = 600\,kPa$. $V_2 = 5V_1$ $T_1 = 300.15K$, 绝热容器.

∵向真空中膨胀.

∴ $W = 0$

∵绝热膨胀

∴ $Q = 0$

∵ $Q = W + \Delta U$

∴ $\Delta U = 0$, ∴ $T_2 = T_1 = 300.15K = 27°C$

由 $PV = nRT$, 得

$P_2/P_1 = V_1/(V_1+V_2) = \cancel{\frac{1}{5}}\ \frac{1}{6}$

∴ $P_2 = \frac{1}{6}P_1 = 100\,kPa$.

3-9 $P_0 = 500\,kPa$. $T_0 = 298.15K$ (25), $P_1 = 50\,kPa$, $T_1 = 283.15K$ (10), 绝热充气.

∵绝热充气 ∴ $\delta Q = 0$ $\delta m_2 = 0$ $\delta W_s = 0$

由开口系统的能量方程, 得

$\delta m_1 \cdot h_1 = dE_{CV} = d(mu)_{CV}$

∴ $\int_0^{m_1} \delta m_1 \cdot h_1 = \int_1^2 d(mu)_{CV}$

$m_1 \cdot h_1 = (mu)_{CV2} - (mu)_{CV1}$ ①

$P_2 = P_0 = 500\,kPa$

$\begin{cases} P_1 V = m_{CV1} R T_1 \\ P_2 V = m_{CV2} R T_2 \end{cases} \Rightarrow \begin{cases} m_{CV1} = \frac{P_1 V}{R T_1} & ② \\ m_{CV2} = \frac{P_2 V}{R T_2} & ③ \end{cases}$

$m_1 = m_{CV2} - m_{CV1}$ ④

将②.③.④代入①中.

$\left(-\frac{P_1 V}{R T_1} + \frac{P_2 V}{R T_2}\right) c_p \cdot T_0 = \frac{P_2 V}{R T_2} \cdot c_V \cdot T_2 - \frac{P_1 V}{R T_1} c_V T_1$

$T_2 = P_2 \cdot \frac{T_0 T_1 k}{P_1 T_0 k + T_1(P_2 - P_1)} = 398.31(K) \approx 125(°C)$

3-12 $p_0 = 1MPa$, $T_0 = 473.15K$, 绝热充气.

(1). $p_1 = 0$, 即容器是真空的

∵绝热充气, ∴ $\delta Q = 0$ $\delta m_2 = 0$, $\delta W_s = 0$

由开口系统的能量方程, 得

$$\delta m_1 \cdot h_1 = dE_{CV} = d(mu)_{CV}$$

$$\int_0^{m_1} \delta m_1 h_1 = \int_1^2 d(mu)_{CV}$$

$$m_1 \cdot h_1 = (mu)_{CV,2} - (mu)_{CV,1}$$

∵ 真空, ∴ $(mu)_{CV,1} = 0$, $m_1 = m_{CV,2}$

∴ $h_1 = u_{CV,2}$

$h_1 = \cancel{\ } c_p \cdot T_0$, $u_{CV,2} = c_V \cdot T_2$

∴ $T_2 = (c_p/c_V) \cdot T_0 = 1.4 \times 473.15 = 662.41 (K) \approx 389 (°C)$.

(2). $\delta m_1 \cdot h_1 = dE_{CV} = d(mu)_{CV} + \delta W$

$$\int_0^{m_1} \delta m_1 h_1 = \int_1^2 d(mu)_{CV} + \int_1^2 \delta W$$

设 $dL = k dp$ $\int_1^2 \delta W = \int pA\,dL = \int pAk\,dp = \frac{1}{2}p^2 A \cdot k = \frac{1}{2}pA \cdot kp = \frac{1}{2}pAL = \frac{1}{2}p_0 V = \frac{1}{2}mRT_2$

∴ $m_1 h_1 = (mu)_{CV_2} - (mu)_{CV_1} + \frac{1}{2}mRT$

$$m_1 h_1 = m_{CV_2} \cdot u_{CV_2} + \frac{1}{2}RT_2 \cdot m_{CV_2}$$

∵ $m_{CV_2} = m_1$

∴ $h_1 = u_{CV_2} + \frac{1}{2}RT_2$

$$c_p \cdot T_0 = c_V \cdot T_2 + \frac{1}{2}RT_2$$

$$T_2 = \frac{c_p T_0}{c_V + \frac{1}{2}R} = \frac{3.5 \cdot T_0}{2.5 + 0.5} = 551.84 (K) \approx 279 (°C)$$

(3). $\int_1^2 \delta W = p_0 \cdot V = m_{CV_2} R T_2$

$$m_1 h_1 = m_{CV_2} \cdot u_{CV_2} + m_{CV_2} R T_2$$

$$m_{CV_2} = m_1$$

∴ $h_1 = u_{CV_2} + RT_2$

$$\therefore T_2 = \frac{C_p \cdot T_0}{c_V + R} = T_0 = 200^\circ C.$$

3-13 $m=1kg$, $p_1 = 800kPa$ $t_1 = 900^\circ C$. $p_2 = 120kPa$ $t_2 = 600^\circ C$ 绝热膨胀.

(1). 定值比热容.

$$\Delta U = mC_V \cdot \Delta T = \frac{7}{2} R_0 \frac{1}{M}(t_2 - t_1)$$

$$= 3.5 \times 8.314 \times \frac{1}{44} \times (-300)$$

$$= -198.402 \ (kJ \ \cancel{每kg})$$

$$\Delta H = mC_p \cdot \Delta T = \frac{9}{2} R \cdot \frac{1}{M}(t_2 - t_1)$$

$$= 4.5 \times 8.314 \times \frac{1}{44} \times (-300)$$

$$= -255.089 \ (kJ \ \cancel{每kg}).$$

(2). 平均比热容.

$$\Delta H = C_p\Big|_{t_1}^{t_2} \Delta T = mC_p\Big|_0^{t_2} t_2 - mC_V\Big|_0^{t_1} \cdot t_1$$

$$= 600 \times 1.040 - 900 \times 1.104 \ \cancel{\cdots}$$

$$= -369.600 (kJ) \quad \cancel{= -315.779 \ (kJ/kg)} \quad \cancel{= -307.050 (kJ)}$$

$$\Delta U = C_V\Big|_{t_1}^{t_2} \Delta T = m\left(C_p\Big|_0^{t_2} - \frac{R_0}{M}\right) \times t_2 - m\left(C_p\Big|_0^{t_1} - \frac{R_0}{M}\right) \times t_1$$

$$\cancel{= -249.104 \ (kJ/kg)} \quad \cancel{= -250.364 (kJ)}$$

$$= -312.914 (kJ)$$

(3). $Q = \Delta H + W_t$

$\because Q = 0$

$\therefore W_t = -\Delta H$

按定值比热容 $W_t = -\Delta H = 255.089 \ (kJ)$

按平均比热容 $W_t = -\Delta H = \cancel{315.779 (kJ)}$ $\cancel{307.050}$ $369.600 (kJ)$

3-17 $t_1=300°C$ $\dot m_1=120kg/h$ $t_2=200°C$, $\dot m_2=210kg/h$，混合前后压力相等

$$\dot m_1 h_1+\dot m_2 h_2=(\dot m_1+\dot m_2)\cdot h$$

$$\dot m_1 c_p\cdot t_1+\dot m_2 c_p t_2=(\dot m_1+\dot m_2)\cdot c_p t$$

$$t=\frac{\dot m_1 t_1+\dot m_2 t_2}{\dot m_1+\dot m_2}$$

$$=582.24(K)\approx 309(°C)$$

3-19 $m=1kg$. $V=0.03m^3$ $P=2068.4kPa$ ~~t₂~~ $T_2=2T_1$ 定压膨胀

$c_p=1.01KJ(kg\cdot K)$ $R=0.287KJ(kg\cdot K)$

$$PV=mRT_1$$

$$\therefore T_1=\frac{PV}{mR}=\frac{2068.4\times0.03}{0.287}=216.209(K)$$

~~ΔH=m.~~

$$\Delta H=m\cdot c_p\cdot(T_2-T_1)$$

$$=1\times1.01\times216.209$$

$$=218.371(kJ)$$

$$\Delta U=m\cdot c_V\cdot(T_2-T_1)$$

$$=m\cdot(c_p-R)\cdot(T_2-T_1)$$

$$=1\times(1.01-0.287)\times216.209$$

$$=\cancel{157.882}\ 156.319(kJ)$$

∵等压过程

$$\therefore Q_p=\Delta H=218.371(kJ)$$

$$Q_p=\Delta U+W$$

$$W=Q_p-\Delta U$$

$$=\cancel{60.489}\ 62.052(kJ)$$

3-20. $A=100cm^2$ $H=10cm$ $m=195kg$ $P_0=771mmHg$

$T=27°C$ $\Delta m=-100kg$ ，求$\Delta h$，$Q$

∵充分换热

∴$\Delta T=0$ ∴$\Delta U=0$.

∵$Q=\Delta U+W$ ∴$Q=W$ ①

$P_0=771\times13.6\times9.8=102.759(kPa)$

$P_1=mg/A=195\times9.8/(100\times10^{-4})=191.1(kPa)$

$P_2=(m+\Delta m)g/A=95\times9.8/(100\times10^{-4})=93.1(kPa)$

~~由活塞内外压力平衡，得~~

由$PV=nRT$，T为恒值，得

$(P_0+P_1)H\cdot A=(P_0+P_2)\cdot(H+\Delta h)\cdot A$

$\frac{102.759+191.1}{102.759+93.1}=\frac{10+\Delta h}{10}$

$\Delta h=5(cm)$

$W=P_{amb}\cdot\Delta V$

$=(P_0+P_2)\cdot A\cdot\Delta h$

$=(102.759+93.1)\times100\times5\times10^{-6}\times10^3$

$=$ ~~97.929(J)~~ $97.930(J)$

由①，$Q=W=$ ~~97.929(J)~~ $97.930(J)$

10.21

11.3

5-1 $T_1=600°C$ $T_2=40°C$ $Q_1=100kJ\cdot s^{-1}$

(1). $\eta=1-\frac{T_2}{T_1}=1-\frac{313.15}{873.15}=64.136\%$

(2). $\eta=\frac{Q_1-Q_2}{Q_1}=\frac{W}{Q_1}$

∴$W=\eta\cdot Q_1=64.136\ kJ\cdot s^{-1}$

∴$W=64.136\ kW$

10.20

4-2 $m=1kg$. $p_1=0.5MPa$. $p_2=0.1MPa$. $T_1=423.15K$.

(1). 可逆绝热膨胀.

$\because \frac{T}{p^{\frac{k-1}{k}}}=C$. $\therefore \frac{T_1}{T_2}=\left(\frac{p_1}{p_2}\right)^{\frac{k-1}{k}}$

$T_2=\frac{423.15}{5^{2/7}}=267.17K$. $T_2=267$

$W=\Delta U=-\frac{m}{M}C_{v,m}(T_2-T_1)$

$=-1\times\frac{1}{29}\times2.5\times8.314\times(267.17-423.15)$

$=111.79(kJ)$

$\Delta S=\frac{m}{M}\cdot C_{p,m}\ln\frac{T_2}{T_1}-\frac{m}{M}R_0\ln\frac{p_2}{p_1}=0$

(2). 不可逆绝热膨胀.

$W = -\Delta U = -\frac{m}{M} C_{v,m} (T_2 - T_1)$

$= -1 \times \frac{1}{29} \times 2.5 \times 8.314 \times (300 - 423.15)$

$= 88.26 (kJ).$

$T_2 = 3$

$\Delta S = \frac{m}{M} C_{p,m} \ln \frac{T_2}{T_1} - \frac{m}{M} R_0 \ln \frac{p_2}{p_1}$

$= 1 \times \frac{1}{29} \times 3.5 \times 8.314 \times \ln \frac{300}{423.15} - \frac{1}{29} \times 8.314 \times \ln \frac{0.1}{0.5}$

$= 116.29 (J/K)$

(3). 可逆等温膨胀.

$W = \int p dv = \int p d(\frac{mRT}{p}) = \int p (-\frac{mRT}{p^2}) dp = -mRT \ln \frac{p_2}{p_1}$

$\therefore W = -1 \times \frac{1}{29} \times 423.15 \times \ln \frac{0.1}{0.5} \times 8.314$

$= 195.52 (kJ)$

$T_2 = 42$

$\Delta S = -\frac{m}{M} R_0 \ln \frac{p_2}{p_1}$

$= -1 \times \frac{1}{29} \times 8.314 \times \ln \frac{0.1}{0.5}$

$= 461.41 (J/K)$

(4). 可逆 $n=2$.

$\frac{T}{p^{\frac{n-1}{n}}} = C \quad \therefore \frac{T_1}{T_2} = (\frac{p_1}{p_2})^{\frac{1}{2}}$

$T_2 = 13$

$T_2 = \frac{423.15}{\sqrt{5}} = 189.24 (k).$

$W = -\Delta U = -\frac{m}{M} C_v (T_2 - T_1)$

$= -1 \times \frac{1}{29} \times \frac{5}{2} \times 8.314 \times (189.24 - 423.15)$

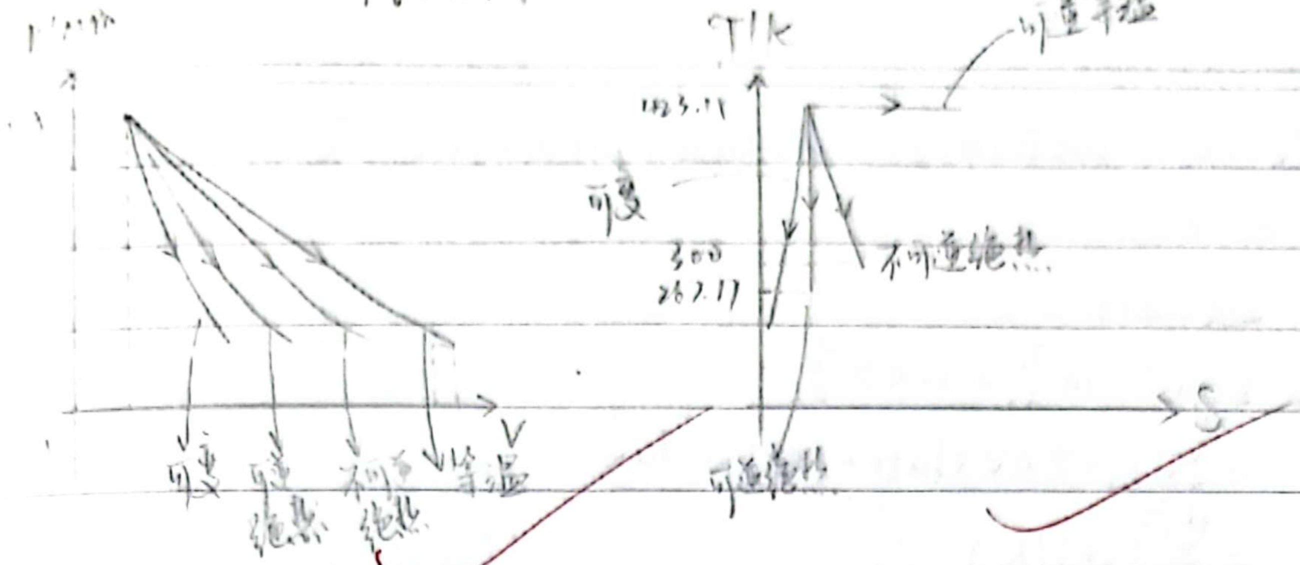
$= 67.06 (kJ)$

$\Delta S = \frac{m}{M} C_{p,m} \ln \frac{T_2}{T_1} - \frac{m}{M} R_0 \ln \frac{p_2}{p_1}$

$= \frac{1}{29} \times 3.5 \times 8.314 \times \ln \frac{189.24}{423.15} - \frac{1}{29} \times 8.314 \times \ln \frac{0.1}{0.5}$

$= -346.05 (J/K).$

$PV=mRT$




4-3 $n=1\,kmol$, $V_1=1m^3$ $V_2=10m^3$ $T=100°C$

(1). 可逆定温.

$W=\int pdV = \int \frac{nRT}{V}dV = nRT\ln\frac{V_2}{V_1}$

$W=1\times10^3\times8.314\times373.15\times\ln10$

$=7143.47\,(kJ)$

$\Delta S = n\cdot R_0 \ln\frac{V_2}{V_1}$

$=1\times8.314\times\ln10$

$=19.14\,(kJ/K)$.

(2). 真空膨胀.

$W=0$

$\Delta S = n\cdot R\cdot\ln\frac{V_2}{V_1}$

$=1\times8.314\times\ln10$

$=19.14\,(kJ/K)$

4-4 $m=5kg$, $O_2$, $T=30°C$. $V_1=3m^3$ $V_2=0.6m^3$

$W=\int pdV = \int\frac{mRT}{V}dV = mRT\ln\frac{V_2}{V_1}$

$W=5\times\frac{1}{32}\times8.314\times303.15\times\ln\frac{1}{5}$

$=-633.81\,(kJ)$

$Q = W = -633.81(kJ)$. ∴放出了633.81kJ的热量

∵ $\Delta T = 0$

∴ $\Delta U = \Delta H = 0$

$\Delta S = m \cdot C_v \cdot \ln\frac{T_2}{T_1} + mR\ln\frac{V_2}{V_1}$

$= \cancel{5 \times \frac{1}{32} \times 8.314 \times \ln 10} + 5 \times 8.314 \times \ln\frac{1}{5}$

$= \cancel{2.99}\ -66.90\ (kJ/K)$ ×

4-5 $B = 101.3kPa$ $T_1 = 13°C$ $\Delta P = 0.1MPa = 100kPa$

∵ $PV = nRT$

∴ $\frac{P_2}{P_1} = \frac{T_2}{T_1}$

∴ $T_2 = T_1 \cdot \frac{B + \Delta P}{B}$

$= 286.15 \times \frac{101.3 + 100}{101.3}$

$= 568.63(K)$

$\Delta u = \cdot C_v \cdot (T_2 - T_1)$

$= \frac{1}{29} \times 2.5 \times 8.314 \times (568.63 - 286.15)$

$= 202.46\ (kJ/kg)$

$\Delta h = C_p \cdot (T_2 - T_1)$

$= \frac{1}{29} \times 3.5 \times 8.314 \times (568.63 - 268.15)$

$= 283.44\ (kJ/kg)$

$\Delta S = C_v \cdot \ln\frac{T_2}{T_1}$

$= \frac{1}{29} \times 2.5 \times 8.314 \times \ln\frac{\cancel{568.63}\ 568.63}{\cancel{88.15}\ 286.15}$

$= 492.18\ (J/(k \cdot kg))$

4-6 $m=6kg$ $p_1=0.3MPa$. $t_1=30°C$ $P_2=0.1MPa$

(1). 定温

$t_2=t_1=30°C$

$$W=mRT\ln\frac{p_1}{p_2}$$

$$=6\times\frac{1}{29}\times8.314\times\ln\frac{0.3}{0.1}\times303.15$$

$$=572.88(kJ)$$

$$Q=W=572.88(kJ)$$

(2). 定熵（可逆绝热）

$$\because T/p^{\frac{k-1}{k}}=C$$

$$\therefore t_2=t_1/\left(\frac{p_1}{p_2}\right)^{\frac{k-1}{k}}$$

$$=303.15/3^{\frac{2}{7}}$$

$$=221.48(K).$$

$$W=m\frac{RT_1}{k-1}\times\left(1-\left(\frac{p_2}{p_1}\right)^{\frac{k-1}{k}}\right)$$

$$=6\times\frac{8.314\times303.15}{0.4}\times\frac{1}{29}\times\left(1-\left(\frac{1}{3}\right)^{\frac{2}{7}}\right)$$

$$=351.20(kJ)$$

$$Q=0$$

(3). $n=1.2$, 多变.

$$t_2=t_1/\left(\frac{p_1}{p_2}\right)^{\frac{n-1}{n}}$$

$$=303.15/3^{\frac{1}{6}}$$

$$=252.43(K).$$

$$W=m\frac{RT_1}{n-1}\times\left(1-\left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}}\right)$$

$$=6\times\frac{8.314\times303.15}{0.2}\times\frac{1}{29}\times\left(1-\left(\frac{1}{3}\right)^{\frac{1}{6}}\right)$$

$$=436.24(kJ)$$

$$\Delta U=m\cdot c_v\cdot(t_2-t_1)$$

$= 6 \times 2.5 \times 8.314 \times \frac{1}{29} \times (252.43 - 303.15)$

$= -218.11 \,(kJ)$

$Q = W + \Delta U$

$= 436.24 - 218.11$

$= 218.13 \,(kJ)$

4-7 解 $P_1 = 0.6MPa$. $v_1 = 0.236 m^3/kg$ $P_2 = 0.12 MPa$. $v_2 = 0.815 m^3/kg$

$\because PV^n = C$

$\therefore \frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^n$

$n = \frac{\ln(V_1/V_2)}{\ln(P_2/P_1)} = \frac{\ln(0.236/0.815)}{\ln(0.12/0.6)} = 1.3$

$W = \frac{RT_1}{n-1}\left(1 - \left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}}\right)$

$= \frac{P_1 V_1}{n-1}\left(1 - \left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}}\right)$

$= \frac{0.6 \times 10^3 \times 0.236}{0.3} \times \left(1 - 0.2^{\frac{0.3}{1.3}}\right)$

$= 146.43 \,(kJ/kg)$

$\Delta U = C_V (T_2 - T_1)$

$= 2.5R\left(\frac{P_2 V_2}{R} - \frac{P_1 V_1}{R}\right)$

$= 2.5 \times (120 \times 0.815 - 600 \times 0.236)$

$= -109.50 \,(kJ/kg)$

$\Delta h = C_p (T_2 - T_1)$

$= 3.5 \times (120 \times 0.815 - 600 \times 0.236)$

$= -153.3 \,(kJ/kg)$

$q = \Delta U + W = -109.50 + 146.43 = 36.93 \,(kJ/kg)$

$\Delta S = c_p \ln \frac{V_2}{V_1} + c_v \ln \frac{P_2}{P_1}$

$$= 3.5\times 8.314\times \frac{1}{29}\times \ln\frac{0.815}{0.236} + 2.5\times 8.314\times \frac{1}{29}\times \ln\frac{0.12}{0.6}$$

$$= 90.06\ (J\cdot K^{-1}\cdot kg^{-1})$$

解.

4-8 $m = 1kg$, $T_1 = 400°C$, $T_2 = 100°C$ $P_2 = \frac{1}{6}P_1$ $W = 200kJ$ $Q = 40kJ$

$$\cancel{\frac{T_1}{T_2} = \left(\frac{P_1}{P_2}\right)^{\frac{n-1}{n}}}$$

$$\cancel{\frac{n-1}{n} = \frac{\ln(T_1/T_2)}{\ln(P_1/P_2)} = \frac{\ln 4}{\ln 6}}$$

$$\cancel{\therefore n = 1/\left(1-\frac{\ln 4}{\ln 6}\right)}$$

$$\Delta U = Q - W$$

$$= 40 - 200$$

$$= -160\ (kJ)$$

$$\Delta U = m c_V (T_2 - T_1)$$

$$-160 = 1\times C_V\times(-300)$$

$$C_V = 533.33\ (J\cdot kg^{-1}\cdot K^{-1})$$

$$\frac{T_1}{T_2} = \left(\frac{P_1}{P_2}\right)^{\frac{n-1}{n}}$$

$$n = 1/\left(1-\frac{\ln(T_1/T_2)}{\ln(P_1/P_2)}\right)$$

$$= 1/\left(1-\frac{\ln(673.15/373.15)}{\ln 6}\right)$$

$$= 1.49$$

$$W = \frac{mRT_1}{n-1}\left(1-\left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}}\right)$$

$$200 = \frac{R\cdot 673.15}{0.49}\times\left(1-\frac{1}{6}^{\frac{0.49}{1.49}}\right)$$

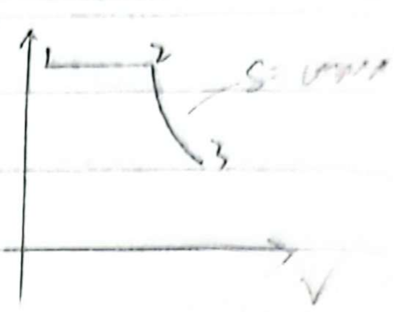
$$\therefore R = 326.97\ (J\cdot kg^{-1}\cdot K^{-1})$$

$$\therefore C_p = C_V + R = 533.33 + 326.97 = 860.30\ (J\cdot kg^{-1}\cdot K^{-1})$$

773.15K

4-10　$m=1kg$.　$N_2$.　$t_1=500°C$　$V_2=0.25\,m^3/kg$

$P_3=0.1\,MPa$.　$V_3=1.73\,m^3/kg$




~~$P_1V_1 = mRt_1$~~

绝热 $PV^k = C$

$\therefore P_2V_2^{1.4} = P_3V_3^{1.4}$

$P_2 = P_3 \cdot \left(\frac{V_3}{V_2}\right)^{1.4}$

$= 0.1 \times \left(\frac{1.73}{0.25}\right)^{1.4}$

$= 1.5\ (MPa)$　　$P_1 = P_2 = 1.5\ (MPa)$

$P_2V_2 = mRt_2$

$t_2 = \frac{P_2V_2}{mR}$

$= \frac{1.5\times10^3\times0.25}{1\times8.314\times\frac{1}{28}}$　$= 1262.93\ (K)$

~~$= 183.70\ (K)$~~

$TV^{k-1} = C$

$t_3 = t_2 \cdot \left(\frac{V_2}{V_3}\right)^{k-1}$

$= 1262.93 \times \left(\frac{0.25}{1.73}\right)^{0.4}$

$= 582.55\ (K)$

$\therefore V_1 = V_2 \cdot (t_1/t_2)$

$= 0.25 \times (773.15/1262.93)$

$= 0.153\ (m^3/kg)$

$\Delta U_{12} = m \cdot C_V \cdot (t_2 - t_1)$

$= 1\times2.5\times8.314\times(1262.93-773.15)\times\frac{1}{28}$

$= 363.57\ (kJ)$

$\Delta U_{23} = m \cdot C_V \cdot (t_3 - t_2)$

$= 1\times2.5\times8.314\times\frac{1}{28}\times(582.55-1262.93)$

$= -505.06\ (kJ)$

$\frac{1000}{22.4} \times 29 = 1.29\,kg$

$W_{12} = P_2(V_2 - V_1)$

$= 1.5\times10^3 \times (0.25 - 0.153)$

$= 145.5\,(kJ)$

$W_{23} = -\Delta U_{23}$

$= 505.06\,(kJ)$

$PV = mRT$

$0.6\times10^6 \times 1 = m \times 8.314 \times \frac{1}{29} \times 573.15$

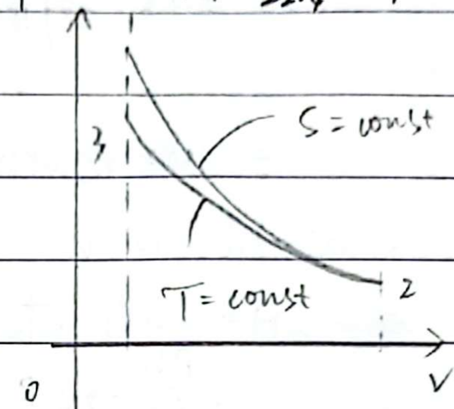
$m = 3.6515\,kg$

$m = \frac{1000}{22.4} \times 29 = 1.29\,kg$

4-11 $P_1 = 0.6\,MPa$. $t_1 = 300\,^\circ C$ $V_2 = 3V_1$

$V_3 = V_1$.

空气




等熵，$PV^k = C$

$P_2 = P_1\left(\frac{V_1}{V_2}\right)^k$

$= 0.6\times\left(\frac{1}{3}\right)^{1.4}$

$= 0.129\,(MPa)$

等温 $P_2V_2 = P_3V_3$

$P_3 = P_2V_2 \cdot \frac{1}{V_3}$

$= 0.129\times3$

$= 0.387\,(MPa)$

等熵. $TV^{k-1} = c$

$\therefore t_2 = t_1\cdot\left(\frac{V_1}{V_2}\right)^{k-1}$

$= 573.15\times\left(\frac{1}{3}\right)^{0.4}$

$= 369.33\,(K)$

等温 $t_3 = t_2 = 369.33\,(K)$

$P_1V_1 = Rt_1$

$V_1 = Rt_1/P_1$

$= 8.314\times\frac{1}{29}\times573.15\times\frac{1}{0.6\times10^6}$

$= 0.274\,(m^3/kg)$

$W = \frac{Rt_1}{k-1}\left(1-\left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}}\right) + Rt_2\ln\frac{V_3}{V_2}$

$= \frac{8.314\times573.15}{0.4}\times\frac{1}{29}\times\left(1-\frac{0.129}{0.6}^{\frac{0.4}{1.4}}\right)+8.314\times\frac{1}{29}\times369.33\times\ln\frac{1}{3}$

$= 146.008$ $\quad -116.32$

$= 29.68\,(kJ/kg)$

$W = mw = 29.68\times1.29 = 38.8\,(kJ)$

$w = \frac{R}{k-1}\,t_1 - t_2$

$= \frac{8.314}{0.4}\times173.15\times\frac{1}{29} = \cancel{146.7}\ 146.08$

$V_2 = 3V_1 = 0.822\,(m^3/kg)$

$V_3 = V_1 = 0.274\,(m^3/kg)$

4-16 $P_1=0.1MPa$. $t_1=16^\circ C$ $V=400m^3/min$

$P_2=0.5MPa$. $t_2=75^\circ C$，可逆

(1). $t_1/t_2=\left(\frac{P_1}{P_2}\right)^{\frac{n-1}{n}}$

$\therefore n=\frac{1}{1-\frac{\ln(t_1/t_2)}{\ln(P_1/P_2)}}=\frac{1}{1-\frac{\ln(289.15/348.15)}{\ln(0.1/0.5)}}=1.13$

$W_{s,n}=\frac{n}{n-1}mR(T_1-T_2)$

$=\frac{1.13}{0.13}\times\frac{0.1\times10^3\times400}{289.15\times\frac{8.314}{29}}\times\frac{8.314}{29}\times(16-75)\times\frac{1}{60}$

$=-1182.42\ (kW)$

$\therefore W=1182.42\ kW$

(2). $\Delta H=mC_p(T_2-T_1)$

$=\frac{P_1V_1}{Rt_1}\cdot3.5R(t_2-t_1)$

$=\frac{0.1\times10^3\times400}{289.15}\times3.5\times(-16+75)$

$=28566\ (kJ/min)$

$Q=\Delta H+W_s$

$=28566-70945$

$=-42379\ (kJ/min)$

4-17 $c=6\%$ $P_1=0.1MPa$. $P_2=0.5MPa$.

$n_1=1.4$ $n_2=1.25$ $n_3=1$

$\lambda_V=\cancel{c\left[1-\left(\frac{P_2}{P_1}\right)^n\right]}=1-c\left[\left(\frac{P_2}{P_1}\right)^{\frac{1}{n}}-1\right]$

$\cancel{=1-c\left[1-\left(\frac{P_1}{P_2}\right)^n\right]}$

~~$\lambda_{V1} = 1 - 6\% \times \left[1 - \left(\frac{1}{5}\right)^{1.4}\right] = 94.6\%$~~

~~$\lambda_{V2} = 1 - 6\% \times \left[1 - \left(\frac{1}{5}\right)^{1.25}\right] = 94.8\%$~~

~~$\lambda_{V3} = 1 - 6\% \times \left[1 - \left(\frac{1}{5}\right)^{1}\right] = 95.2\%$~~

$$\lambda_{V1} = 1 - 6\% \left[5^{\frac{1}{1.4}} - 1\right] = 87.1\%$$

$$\lambda_{V2} = 1 - 6\% \left[5^{\frac{1}{1.25}} - 1\right] = 84.3\%$$

$$\lambda_{V3} = 1 - 6\% \left[5^{\frac{1}{1}} - 1\right] = 76\%$$

11.1

过①, $Q = W = $ ~~[illegible]~~ 11.750(J))

11.3

5-1 $T_1 = 600\,°C$ $T_2 = 40\,°C$ $Q_1 = 100\,kJ \cdot s^{-1}$

(1). $\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{313.15}{873.15} = 64.136\%$

(2). $\eta = \frac{Q_1 - Q_2}{Q_1} = \frac{W}{Q_1}$

$\therefore W = \eta \cdot Q_1 = 64.136\,kJ \cdot s^{-1}$

$\therefore W = 64.136\,kW$

(3). $\eta = 1-\frac{Q_2}{Q_1}$

$Q_2 = (1-\eta)\cdot Q_1$

$= (1-64.136\%)\times 100$

$= 35.864\ (kJ\cdot s^{-1})$

5-2 $T_1 = 1000K$ $T_2 = 400K$, $Q_1 = 1000kJ$ $W = 700kJ$

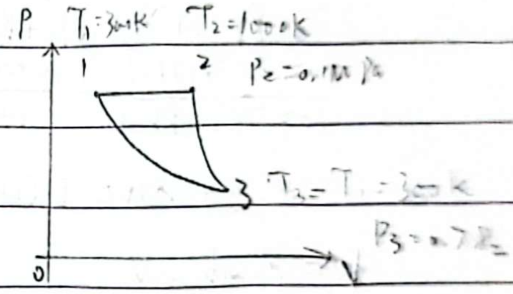
$\eta_c = 1-\frac{T_2}{T_1} = 1-\frac{400}{1000} = 0.6$

$\eta = \cancel{1-}\frac{W}{Q_1} = \frac{700}{1000} = 0.7$

$\eta > \eta_c$

∴这样的循环发动机不可实现.

$P_1 = 0.1MPa$ $pV = mRT$




5-3 $T_1 = \overset{300}{\cancel{1000}}K$. $T_2 = \overset{1000}{\cancel{300}}K$ $p_1 = 0.1MPa$

$m = 1kg$, $c_p = 1.01\ kJ(kg\cdot K)$

$Q_{12} = m\cdot c_p\cdot(T_2-T_1)$

$= 1\times1.01\times(1000-300)$

$= 707(kJ)$

$Q_{23} = 0$

$Q_{31} = mW_{31}$

$= \int_3^1 p\,dV = \int_3^1 \frac{mRT}{V}dV = mRT_1\ln\frac{V_1}{V_3} = mRT_1\ln\frac{p_3}{p_1}$

$\because \frac{T}{p^{\frac{k-1}{k}}} = C$

$\therefore T_2/T_3 = (p_2/p_3)^{\frac{k-1}{k}}$

$\therefore p_3/p_2 = \left(\frac{1000}{300}\right)^{-\frac{1.4}{0.4}} = 1.479\times10^{-2}$

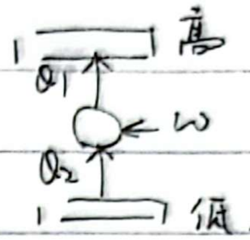
$Q_{31} = mRT_1\ln\frac{p_3}{p_1} = mRT_1\ln\frac{p_3}{p_2}$

$= 1\times\frac{8.314}{29}\times300\times\ln(1.479\times10^{-2})$

$= -362.416\ (kJ)$

$$W = Q_{12} + Q_{31}$$
$$= 707 - 362.416$$
$$= 344.584\ (kJ)$$
$$\eta = \frac{W}{Q_{12}} = \frac{344.584}{707} = 48.739\%$$

5-5 $Q_1 = 1\times10^5\ kJ/h$ $T_1 = 293.15K$ $T_2 = 263.15K$




(1). $\eta = \cancel{\frac{T_1}{T_2}}\ 1 - \frac{T_2}{T_1} = 1 - \frac{Q_2}{Q_1}$

$$1 - \frac{263.15}{293.15} = 1 - \frac{Q_2}{1\times10^5}$$
$$Q_2 = 0.898\times10^5\ (kJ/h)$$
$$= 89766\ (kJ/h)$$

(2). $W = Q_1 - Q_2$
$$= 100000 - 89766$$
$$= 10234\ (kJ/h)$$
$$= 2.843\ (kW)$$

(3). $W = Q_1$
$$= 1\times10^5\ kJ/h$$
$$= 27.778\ (kW)$$

5-6 $T_1 = 293.15K$ $T_2 = 273.15K$

(1). $Q_1 = 20\times1200$
$$= 24000\ (kJ/h)$$

$$W = Q_1 - Q_2$$
$$= Q_1 - (1-\eta)Q_1$$
$$= \eta Q_1 = \left(1 - \frac{T_2}{T_1}\right)\cdot Q_1 = \left(1 - \frac{273.15}{293.15}\right)\times24000$$
$$= 1637.39\ (kJ/h) = 0.455\ (kW)$$

(2). ~~设温差为t~~

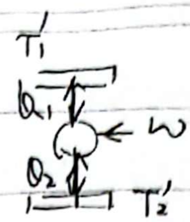
~~$W = \eta = 1 - \frac{T_2}{T_1}$~~

~~$= 1 - \frac{T_2'}{T_1'} = 1 - \frac{T_2}{T_1}$~~

~~$T_1' = \frac{T_1}{T_2} \cdot T_2'$~~ ~~$T_1' = \frac{T_1}{T_2} \cdot T_2'$~~

~~$293.15 = \frac{273.15}{293.15} \times T_2'$~~ ~~$= \frac{293.15}{273.15} \times 293.15$~~

~~$T_2' = 41.464(°C)$~~ ~~$= 41.464(°C)$~~




$W \frac{T_2'}{T_1' - T_2'} = 1200(T_1' - T_2')$

5-7

$1637.39 \times 293.15 = 1200(T_1' - 293.15)^2$

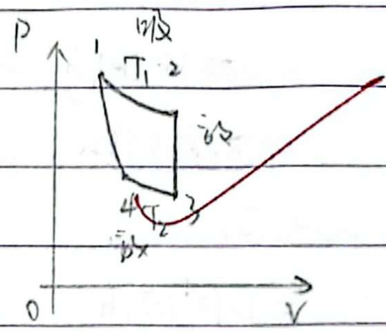
$T_1' = 313.15(K) = 40(°C)$

$Q_b = Q \cdot \eta \cdot \varepsilon_V$

$= 10000 \times 30\% \times 5$

$= 15000 (kJ/h)$

5-14

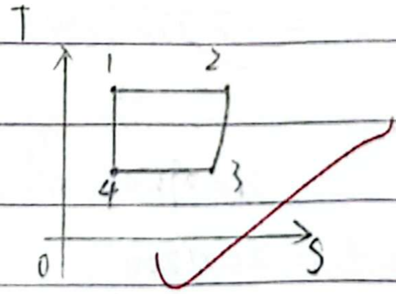



$Q_{12} = T_1(S_2 - S_1) > 0$ $\quad \Delta S_{12} > 0$

$Q_{23} = mC_V \cdot (T_3 - T_1) < 0$ $\quad \Delta S_{23} < 0$

$Q_{34} = T_2(S_4 - S_3) < 0$ $\quad \Delta S_{34} < 0$

$Q_{41} = 0$ $\quad \Delta S_{41} = 0$




$\eta = 1 - \frac{|Q_{23}| + |Q_{34}|}{Q_{12}}$

$= 1 - \frac{C_V(T_1 - T_3) + T_2(S_3 - S_4)}{T_1(S_2 - S_1)}$

5-15 $T_1 = 2000K$ $T_2 = 600K$ $T_1' = 1940K$ $T_2' = 660K$

(1). $\eta = 1 - \frac{T_2'}{T_1'} = 1 - \frac{660}{1940} = 65.979\%$

(2). $\eta_c = 1 - \frac{T_2}{T_1} = 1 - \frac{600}{2000} = 70\%$

$\Delta W = Q(\eta_c - \eta) = 1000 \times (70\% - 65.979\%) = 40.21(kJ)$

5-16 $P_1=0.4MPa$, $T_1=313.15K$ $V_1=V_2=0.1m^3$

$P_2=0.2MPa$, $T_2=293.15K$

~~两~~由 $PV=mRT$

$$m_1=\frac{P_1V_1}{RT_1}=\frac{0.4\times10^6\times0.1}{8.314\times\frac{1}{29}\times313.15}=0.446(kg)$$

$$m_2=\frac{P_2V_2}{RT_2}=\frac{0.2\times10^6\times0.1}{8.314\times\frac{1}{29}\times293.15}=0.238(kg)$$

恒容混合，$\Delta U=0$

$$\Delta U_1+\Delta U_2=0$$

$$m_1\cdot C_V(T-T_1)+m_2C_V(T-T_2)=0$$

$$T=\frac{m_1T_1+m_2T_2}{m_1+m_2}$$

$$=306.191(K)$$

$$\Delta S_1=m_1C_V\ln\frac{T}{T_1}+m_1R\ln\frac{V_1+V_2}{V_1}$$

$$=\left[0.446\times\overset{\times2.5}{8.314}\times\frac{1}{29}\times\ln\frac{306.191}{313.15}+0.446\times8.314\times\frac{1}{29}\times\ln2\right]\times10^3$$

$$=81.445(J\cdot K^{-1})$$

$$\Delta S_2=m_2C_V\ln\frac{T}{T_2}+m_2R\ln\frac{V_1+V_2}{V_1}$$

$$=\left[0.238\times2.5\times8.314\times\frac{1}{29}\times\ln\frac{306.191}{293.15}+0.238\times8.314\times\frac{1}{29}\times\ln2\right]\times10^3$$

$$=54.719(J\cdot K^{-1})$$

$$\Delta S=\Delta S_1+\Delta S_2$$

$$=81.445+54.719$$

$$=136.164(J\cdot K^{-1})$$

5-19 $P_1=0.1MPa$ $t_1=303.15K$ $P_2=1MPa$ $n=1.3$ $T_0=290K$ $m=1kg$

$\because \frac{t}{p^{\frac{k-1}{k}}}=C$

$\therefore t_2=t_1\cdot\left(\frac{P_2}{P_1}\right)^{\cancel{\frac{k-1}{k}}\frac{n-1}{n}}$

$$=303.15\times10^{\frac{0.3}{1.3}}=\cancel{585.291}\ 515.735\ (K)$$

$\Delta S_{sys} = mC_p \ln\frac{t_2}{t_1} - mR\ln\frac{p_2}{p_1}$

$= [1\times 8.314\times\frac{1}{29}\times 3.5\times\ln\frac{\cancel{588.291}\ 515.735}{303.15} - 1\times 8.314\times\frac{1}{29}\times\ln 10]\times 1000$

$= \cancel{3.607}\ \cancel{(kJ\cdot K^{-1})}$

$= \cancel{-0.127\ (kJ\cdot K^{-1})} = -126.948\ (J\cdot K^{-1})$

$\cancel{W = mR}$

$Q = m\cdot\frac{n-k}{n-1}\cdot C_v(t_2-t_1)$

$= 1\times\frac{-0.1}{0.3}\times 2.5\times 8.314\times\frac{1}{29}\times(515.735-303.15)$

$= \cancel{-42.713\ (kJ)}\ -50.788\ (kJ)$

$\Delta S_{amb} = \frac{-Q}{T_0} = \cancel{\frac{42.713}{290} = 0.147\ (kJ\cdot K^{-1})}$

$= \frac{50.788}{290}\times 1000$

$= \cancel{0.175\ (kJ\cdot K^{-1})} = 175.131\ (J\cdot K^{-1})$

$\Delta S_{iso} = \Delta S_{sys}+\Delta S_{amb}$

$= \cancel{-0.127+0.175} = -126.948+175.131$

$= \cancel{0.048\ (kJ\cdot K^{-1})} = 48.183\ (J\cdot K^{-1})$

11.15

12.03

9-2 $\dot{m}=1kg/s$ $P_1=0.3MPa$ $t_1=200^\circ C$ $C_1=20m/s$ $P_2=0.2MPa$

$P_1V_1=Rt_1$

$0.3\times10^6\times V_1=\frac{1}{29}\times8.314\times473.15$

$V_1=4.52\times10^{-4}(m^3/g)$

$P_1V_1^k=P_2V_2^k$

$V_2^k=V_1^k\frac{P_1}{P_2}$

$V_2=V_1(\frac{P_1}{P_2})^{\frac{1}{k}}$

$=4.52\times10^{-4}\times1.5^{\frac{1}{1.4}}$

$=6.04\times10^{-4}(m^3/g)$

$C_2=\sqrt{2\frac{k}{k-1}P_1V_1[1-(\frac{P_2}{P_1})^{\frac{k-1}{k}}]}$

$=\sqrt{2\times\frac{1.4}{0.4}\times0.3\times10^6\times4.52\times10^{-4}\times(1-(\frac{2}{3})^{\frac{0.4}{1.4}})\times1000}$

$=322.23(m/s)$

$f_2=\frac{\dot{m}V_2}{C_2}$

$=\frac{1\times6.04\times10^{-4}\times1000}{322.23}\times10^4$

$=18.7(cm^2)$

9-5 $P_1=0.7MPa$ $t_1=947^\circ C$ $C_1=0$ $P_2=0.5MPa$ $P_2'=0.12MPa$

$\dot{m}=0.5kg/s$

$C_2=\sqrt{2\frac{k}{k-1}Rt_1[1-(\frac{P_2}{P_1})^{\frac{k-1}{k}}]}$

$=\sqrt{2\times\cancel{0.4}\frac{1.4}{0.4}\times\frac{8314}{29}\times1220.15\times(1-(\frac{5}{7})^{\frac{0.4}{1.4}})}$

$=\cancel{43}473.75(m/s)$

当地声速 $a=\sqrt{kRT}=\cancel{\sqrt{1.4\times\frac{8314}{29}}}$

$=\sqrt{1.4\times8314\times\frac{1}{29}\times\cancel{1220.15}1108.31}$

$=\cancel{699.8}(m/s)$

666.96

$P_1V_1^k=P_2V_2^k$

$PV=RT$

$\frac{T_2}{T_1}=\frac{P_2V_2}{P_1V_1}$

$=\frac{P_2}{P_1}(\frac{P_1}{P_2})^{\frac{1}{k}}$

$=(\frac{P_1}{P_2})^{\frac{1-k}{k}}$

$=(\frac{P_2}{P_1})^{\frac{k-1}{k}}$

$t_2=1108.31$

当$P_2=0.5$ MPa时 $c_2<a$ $M<1$，为渐缩喷管

$$p_1 v_1^k = p_2 v_2^k$$

$$v_2 = v_1\left(\frac{p_1}{p_2}\right)^{\frac{1}{k}}$$

$$= \frac{Rt_1}{p_1}\left(\frac{p_1}{p_2}\right)^{\frac{1}{k}}$$

$$= 8314\times 1220.15\times\frac{1}{29}\times\left(\frac{7}{5}\right)^{\frac{1}{1.4}}\times\frac{1}{0.7\times10^6}$$

$$= 0.635\ (m^3/kg)$$

$$f_2 = \frac{\dot{m}\cdot v_2}{c_2}$$

$$= 0.5\times0.635\times\frac{1}{473.75}\times10^4$$

$$= 6.70\ (cm^2)$$

当$P_2'=0.15$ MPa时.

$$c_2' = \sqrt{2\frac{k}{k-1}Rt_1\left(1-\left(\frac{p_2}{p_1}\right)^{\frac{k-1}{k}}\right)}$$

$$= \sqrt{2\times\frac{1.4}{0.4}\times8314\times\frac{1}{29}\times1220.15\times\left(1-\left(\frac{0.12}{0.7}\right)^{\frac{0.4}{1.4}}\right)}$$

$$= 984.49\ (m/s)$$

$c_2'>a$ $M>1$ ．为渐缩渐扩喷管

$$p_1 v_1^k = p_2 v_2'^k$$

$$v_2' = v_1\left(\frac{p_1}{p_2'}\right)^{\frac{1}{k}}$$

$$= 8314\times\frac{1}{29}\times1220.15\times\frac{1}{0.7\times10^6}\times\left(\frac{0.7}{0.12}\right)^{\frac{1}{1.4}}$$

$$= 1.76\ (m^3/kg)$$

$$f_2' = \frac{\dot{m}\cdot v_2'}{c_2'}$$

$$= 0.5\times1.76\times\frac{1}{984.49}\times10^4$$

$$= 8.93\ (cm^2)$$

$\dot{m}=\frac{cf}{v}$ $f\cdot\frac{mv}{c}$

# 等截面

9-6 $C_1=100m/s$ $p_1=0.15MPa$ $C_2=171.4m/s$ $P_2=0.14MPa$ $f=0.1m^2$

$t_1=100°C$

(1). $p_1v_1=Rt_1$

$0.15\times10^6 v_1=8314\times\frac{1}{29}\times373.15$

$v_1=0.713(m^3/kg)$

$\dot{m}=\frac{c_1f}{v_1}$

$=100\times0.1\times\frac{1}{0.713}$

$=14.0(kg/s)$

(2). $\dot{m}=\frac{c_2f}{v_2}$

$v_2=171.4\times0.1\times\frac{1}{14.0}$

$=1.22(m^3/kg)$

$P_2v_2=Rt_2$

$t_2=0.14\times10^6\times1.22\times\frac{1}{8314}\times29$

$=595.77(k)$

$=322.62(°C)$

(3). $Q=c_p(t_2-t_1)+\frac{1}{2}(c_2^2-c_1^2)$

$=1.01\times222.62+\frac{1}{2}\times(171.4^2-100^2)\times\frac{1}{1000}$

$=234.54(kJ/kg)$

9-11 $P_0=1.0MPa$ $T_0=800K$ 渐缩渐扩喷管，$f_0=20cm^2$

出口 $M=2$

(1). 喉部到临界状态.

$P_c=0.528P_0=0.528(MPa)$

$T_c=\frac{2}{k+1}T_0=\frac{2}{2.4}\times 800=666.67(K)$

$v_c=\frac{RT_c}{P_c}=8314\times\frac{1}{29}\times 666.67\times\frac{1}{0.528}\times 10^{-6}$

$=0.367(m^3/kg)$

$C_c=a=\sqrt{kRT_c}=\sqrt{1.4\times 8314\times\frac{1}{29}\times 666.67}=517.28(m/s)$

(2). $M=2$

$C_2=2a=\cancel{1034.56(m/s)}$ $2\sqrt{kP_2v_2}$

$C_2=\sqrt{2\frac{k}{k-1}P_0v_0[1-(\frac{P_2}{P_0})^{\frac{k-1}{k}}]}$

$\cancel{\therefore 1-(\frac{P_2}{P_0})^{\frac{k-1}{k}}=0.667}$

$\cancel{\therefore P_2=}$

$\overset{2}{\cancel{4k}}P_2v_2=\cancel{2}\frac{\cancel{k}}{k-1}P_0v_0[1-(\frac{P_2}{P_0})^{\frac{k-1}{k}}]$

$1=1.8(\frac{P_2}{P_0})^{\frac{k-1}{k}}$

$P_2=0.128(MPa)$

$\frac{T_2}{T_0}=(\frac{P_2}{P_0})^{\frac{k-1}{k}}$

$T_2=800\times(\frac{0.128}{1})^{\frac{0.4}{1.4}}$

$=444.64(K)$

$P_2v_2=RT_2$

$v_2=\frac{8314}{29}\times 444.64\times\frac{1}{0.128}\times 10^{-6}$

$=0.996(m^3/kg)$

$C_2=2\sqrt{kRT_2}$

$=2\sqrt{1.4\times 8314\times\frac{1}{29}\times 444.64}$

$=844.90(m/s)$

$P_2v_2^k=P_0v_0^k$

$v_2=(\frac{P_0}{P_2})^{\frac{1}{k}}v_0$

$2P_2v_0(\frac{P_0}{P_2})^{\frac{1}{k}}=\frac{1}{k-1}P_0v_0$

$2(\frac{P_0}{P_2})^{\frac{1}{k}-1}=\frac{1}{k-1}[1-$

$2(\frac{P_2}{P_0})^{\frac{k-1}{k}}=\frac{1}{k-1}[1-$

$4.1(\ )=2.1$

$$\cancel{\dot{m} = \frac{f_c v_c}{c_0} = 20 \times 10^{-4} \times 0.362 \times \frac{1}{517.28}}$$

$$\dot{m} = \frac{f_2 c_c}{v_c} = 20 \times 517.28 \times \frac{1}{0.362} \times 10^{-4}$$

$\cancel{= 2.858}$

$= 2.86\ (kg/s)$

$$f_2 = \dot{m} \cdot v_2 / c_2$$

$$= 2.86 \times 0.996 \times \frac{1}{844.90} \times 10^{4}$$

$= 33.71\ (cm^2)$

(3). $\dot{m} = 2.86\ kg/s$

12.20

$\lambda_{V1} = 1 - 6\% \left[5^{\frac{1}{1.4}} - 1\right] = 87.1\%$

$\lambda_{V2} = 1 - 6\% \left[5^{\frac{1}{1.25}} - 1\right] = 84.3\%$

$\lambda_{V3} = 1 - 6\% \left[5^{\frac{1}{1}} - 1\right] = 76\%$

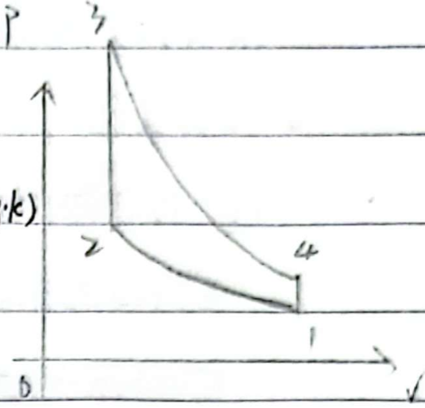
11.1

12.22

10-7 Otto循环

$\varepsilon = 8 \quad T_1 = 300K, \quad P_1 = 97kPa$

$Q_{23} = 700kJ/kg \quad k = 1.41, \quad c_V = 0.73\ kJ/(kg \cdot k)$




$\eta = 1 - \frac{1}{\varepsilon^{k-1}} = 57.4\%$

1-2为绝热压缩

$\cancel{T_2 = T_1 \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}}}$

$\cancel{= 300K}$

$T_2 = T_1 \left(\frac{V_1}{V_2}\right)^{k-1}$

$= 300 \times 8^{0.41}$

$= 703.7(K)$

2-3为定容加热，

$Q_{23} = \Delta U_{23} = c_V(T_3 - T_2)$

$\therefore T_3 = T_2 + \frac{Q_{23}}{c_V}$

$= 1662.6(K)$

∵1-2为绝热

$P_2 = P_1 \left(\frac{V_1}{V_2}\right)^k$

$= 97 \times 8^{1.41}$

$= 1820.24(kPa)$

2-3为定容

$P_3 = P_2 \frac{T_3}{T_2}$

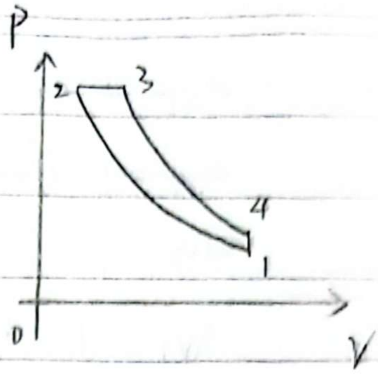
$= 1820.24 \times \frac{1662.6}{703.7}$

$= 4300.6(kPa)$

$W_{sn} = Q_{23} \cdot \eta$

$= 401.8(kJ/kg)$

Disel循环




10-8 $\varepsilon=17$, $T_1=293K$, $p_1=105kPa$

$T_3=1900K$, $c_p=1.02kJ/(kg\cdot K)$, $k=1.41$

(1). 1-2 为绝热

$T_2=T_1\left(\frac{V_1}{V_2}\right)^{k-1}$

$=293\times17^{0.41}$

$=936.16(K)$

$P_2=P_1\left(\frac{V_1}{V_2}\right)^{k}$

$=105\times17^{1.41}$

$=5703.25(kPa)$

2-3 为定压

$P_3=P_2(\cancel{\frac{T_3}{T_2}})=5703.25(kPa)$

$\cancel{=5703.25\times\frac{1900}{936.16}}$

$\cancel{=[illegible](kPa)}$

3-4 为绝热

$T_4=T_3\left(\frac{V_3}{V_4}\right)^{k-1}$

$=T_3\left(\cancel{\frac{P_3}{P_4}}\ \frac{V_3}{V_1}\right)^{k-1}$

$=T_3\left(\frac{T_3}{T_1}\cdot\frac{P_1}{P_3}\right)^{k-1}$

$=1900\times\left(\frac{1900}{293}\times\frac{105}{5703.25}\right)^{0.41}$

$=\cancel{[illegible]}\ 794.89(K)$

4-1 为定容

$P_4=P_1\left(\frac{T_4}{T_1}\right)$

$=105\times\frac{794.89}{293}$

$=\cancel{[illegible]}$
$284.86(kPa)$

(2). $\eta=1-\frac{C_v(T_4-T_1)}{C_p(T_3-T_2)}$

$\cancel{=1-\frac{0.73\times(594.66-293)}{1.02\times(1900-936.16)}}$

$=1-\frac{0.73\times(794.89-293)}{1.02\times(1900-936.16)}$

$=62.73\%$

12.29